Module-III
Interpolation

3 Interpolation

- Finite difference operators: Forward, Backward, Central, Average, Shift and Differential
- Newton's forward difference interpolation
- Newton's backward difference interpolation
- Lagrange interpolation
- Newton's Divided Difference Interpolation
- Stirling's interpolation formula
- Cubic spline interpolation for a tabulated function with equally spaced data

VIT BHOPAL
www.vitbhopal.ac.in

For any given pair of values $(x_k, y_k),\ k = 0, 1, 2, 3, \cdots, n$ with equal-spaced abscissas of a function $y = f(x)$, we defined the *forward difference operator* Δ as follows: The first forward difference is usually expressed as

$$\Delta y_i = y_{i+1} - y_i,\ i = 0, 1, \cdots, (n-1) \tag{1}$$

To be explicit, we write

$$\begin{aligned} \Delta y_0 &= y_1 - y_0 \\ \Delta y_1 &= y_2 - y_1 \\ &\vdots \\ \Delta y_{n-1} &= y_n - y_{n-1} \end{aligned}$$

These differences are called *first differences of the function y* and are denoted by the symbol Δy_i. Here, Δ is called *forward difference operator*.

Similarly, the difference of the first differences are called *second differences,* defined by

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0, \quad \Delta^2 y_1 = \Delta y_2 - \Delta y_1$$

Thus, in general

$$\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i$$

Here Δ^2 is called the *second difference operator*. Thus, continuing, we can define, r^{th} difference of y, as

$$\Delta^r y_t = \Delta^{r-1} y_{r+1} - \Delta^{r-1} y_i \tag{2}$$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
x_0	y_0					
		Δy_0				
x_1	y_1		$\Delta^2 y_0$			
$(= x_0 + h)$		Δy_1		$\Delta^3 y_0$		
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$	
$(= x_0 + 2h)$		Δy_2		$\Delta^3 y_1$		$\Delta^5 y_0$
x_3	y_3		$\Delta^2 y_2$		$\Delta^4 y_1$	
$= (x_0 + 3h)$		Δy_3		$\Delta^3 y_2$		
x_4	y_4		$\Delta^2 y_3$			
$= (x_0 + 4h)$		Δy_4				
x_5	y_5					
$= (x_0 + 5h)$						

Figure 1: Forward difference table

It can be noted that the subscription remains constant along each diagonal of the table. The first term in the table, that is y_0 is called the *leading term*, while the differences $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0, \cdots$ are called leading differences.

Example 3.1.1

Construct a forward difference table for the following values of x and y:

x	0.1	0.3	0.5	0.7	0.9	1.1	1.3
y	0.003	0.067	0.148	0.248	0.370	0.518	0.697

x	y	Δ y	Δ^2 y	Δ^3 y	Δ^4 y	Δ^5 y	Δ^6 y
0.1	0.003						
		0.064					
0.3	0.067		0.017				
		0.081		0.002			
0.5	0.148		0.019		0.001		
		0.1		0.003		0	
0.7	0.248		0.022		0.001		0
		0.122		0.004		0	
0.9	0.37		0.026		0.001		
		0.148		0.005			
1.1	0.518		0.031				
		0.179					
1.3	0.697						

Practice problem

1. Construct a forward difference table for the following set of values:

x_i	0	2	4	6	8	10	12	14
y_i	625	81	1	1	81	625	2401	65611

2. Construct a forward (diagonal) difference table for the following set of values:

x_i	1	2	3	4	5
y_i	4	13	34	73	136

Example 3.2.1

Using Newton's forward formula. find the value of $\sin 52^o$ from the following:

θ^o	45^o	50^o	55^o	60^o
$\sin\theta$	0.7071	0.7660	0.8192	0.8660

Solution:

θ^o	$\sin\theta$	Δ	Δ^2	Δ^3
45^o	0.7071			
		0.0589		
50^o	0.7660		-0.0057	
		0.0532		-0.0007
55^o	0.8192		-0.0064	
		0.0468		
60^o	0.8660			

Applying Newton's forward difference interpolation formula.

$$y_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

Here $y_n(x) = \sin 52^o$

$$y_0 = 0.7071, \Delta y_0 = 0.0589, \Delta^2 y_0 = -0.0057, \Delta^3 y_0 = -0.0007$$

$$h = x_1 - x_0 = 50 - 45 = 5$$

$$p = \frac{x - x_0}{h} = \frac{52-45}{5} = \frac{7}{5} = 1.4$$

$$\sin 52^o = 0.7071 + (1.4)(0.0589) + \frac{1.4(0.4)}{2}(-0.0057)$$

$$+ \frac{1.4(0.4)(-0.6)}{6}(-0.0007)$$

$$= 0.7071 + 0.08246 - 0.0016 + 0.00004$$

$$\sin 52^o = 0.7880$$

Example 3.2.2

Using Newton's forward formula, find the value of $f(218)$ if,

x	100	150	200	250	300	350	400
$f(x)$	10.63	13.03	15.04	16.81	18.42	19.90	21.27

x	$f(x)$	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
100	10.63						
		2.4					
150	13.03		-0.39				
		2.01		0.15			
200	15.04		-0.24		-0.07		
		1.77		0.08		0.02	
250	16.81		-0.16		-0.05		0.02
		1.61		0.03		0.04	
300	18.42		-0.13		-0.01		
		1.48		0.02			
350	19.90		-0.11				
		1.37					
400	21.27						

$$
\begin{aligned}
y_n(x) = y_0 \\
&+ p\Delta y_0 \\
&+ \frac{p(p-1)}{2!}\Delta^2 y_0 \\
&+ \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 \\
&+ \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0 \\
&+ \frac{p(p-1)(p-2)(p-3)(p-4)}{5!}\Delta^5 y_0 \\
&+ \frac{p(p-1)(p-2)(p-3)(p-4)(p-5)}{6!}\Delta^6 y_0
\end{aligned}
$$

Here

$$h = x_1 - x_0 = 150 - 100 = 50$$

$$y_n(x) = f(218), p = \frac{x - x_0}{h} = \frac{218 - 100}{50} = 2.36$$

$$
\begin{aligned}
f(218) &= 10.63 + (2.36)(2.4) \\
&+ \frac{(2.36)(1.36)}{2}(-0.39) \\
&+ \frac{(2.36)(1.36)(0.36)}{6}(-0.39) \\
&+ \frac{(2.36)(1.36)(0.36)(-0.64)}{24}(0.15) \\
&+ \frac{(2.36)(1.36)(0.36)(-0.64)(-1.64)}{120}(-0.07) \\
&+ \frac{(2.36)(1.36)(0.36)(-0.64)(-1.64)(-2.64)}{720}(0.02) \\
&= 10.63 + 5.664 - 0.6259 + 0.0289 + 0.0022 + 0 + 0 \\
f(218) &= 15.6993
\end{aligned}
$$

Definition 3.3.1 (Backward Differences)

The differences $y_1 - y_0, y_2 - y_1, \cdots, y_n - y_{n-1}$ when denoted by $\nabla y_1, \nabla y_2, \cdots, \nabla y_n$, respectively, are called first backward difference. Thus, the first backward differences are $\nabla y_r = y_r - y_{r-1}$. This formula is useful when the value of $f(x)$ is required near the end of the table. h is called the interval of difference and $u = \frac{x - x_n}{h}$, Here an is last term.

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
x_0	y_0					
		∇y_1				
x_1 $(= x_0 + h)$	y_1		$\nabla^2 y_2$			
		∇y_2		$\nabla^3 y_3$		
x_2 $(= x_0 + 2h)$	y_2		$\nabla^2 y_3$		$\nabla^4 y_4$	
		∇y_3		$\nabla^3 y_4$		$\nabla^5 y_5$
x_3 $(= x_0 + 3h)$	y_3		$\nabla^2 y_4$		$\nabla^4 y_5$	
		∇y_4		$\nabla^3 y_5$		
x_4 $(= x_0 + 4h)$	y_4		$\nabla^2 y_5$			
		∇y_5				
x_5 $(= x_0 + 5h)$	y_5					

Figure 2: Backward difference table

Example 3.3.2

Given $\sin 45^o = 0.7071$, $\sin 50^o = 0.7660$, $\sin 55^o = 0.8192$, $\sin 60^o = 0.8660$. Find $\sin 57^o$ using on appropriate interpolation formula.

We can form a table of values as follows

x	45^o	50^o	55^o	60^o
y	0.7071	0.7660	0.8192	0.8660

Since $\sin 57^o$ is closer x_n value, we can choose Newton's backward interpolation formula.

x	y	∇	∇^2	∇^3
45^o	0.7071			
		0.0589		
50^o	0.7660		-0.0057	
		0.0532		-0.0007
55^o	0.8192		-0.0064	
		0.0468		
60^o	0.8660			

Here,

$$r = \frac{x - x_n}{h} = \frac{57 - 60}{5} = -0.6$$

We have newton's backward interpolation formula

$$y_r = y_n + r\nabla y_n + \frac{(r)(r+1)}{2!}\nabla^2 y_n + \frac{(r)(r+1)(r+2)}{3!}\nabla^3 y_n + \cdots$$

$$y_{57} = 0.8660 + (-0.6)(0.0468) + \frac{(-0.6)(-0.6+1)}{2!}(-0.0064)$$

$$+ \frac{(-0.6)(0.6+1)(0.6+2)}{3!}(-0.0007)$$

$$\sin(57) = 0.8387$$

Example 3.3.3

Find the value of $f(17)$ form following table of values

x	0	5	10	15	20
$f(x)$	1.0	1.6	3.8	8.2	15.4

x	$f(x)$	∇y_n	$\nabla^2 y_n$	$\nabla^3 y_n$	$\nabla^4 y_n$
0	1.0				
		6			
5	1.6		1.6		
		2.2		0.6	
10	3.8		2.2		0
		4.4		0.6	
15	8.2		4.4		
		7.2			
20	15.4				

Here,

$$r = \frac{17-20}{5} = -0.6$$

We have newton's backward interpolation formula

$$y_r = y_n + r\nabla y_n + \frac{(r)(r+1)}{2!}\nabla^2 y_n + \frac{(r)(r+1)(r+2)}{3!}\nabla^3 y_n + \cdots$$

$$y_{57} = 15.4 + (-0.6)(7.2) + \frac{(-0.6)(-0.6+1)}{2!}(4.4) + \frac{(-0.6)(-0.6+1)(-0.6+2)}{3!}(0.6) + \frac{(-0.6)(-0.6+1)(-0.6+2)(0.6+3)}{4!}(0)$$

$$f(17) = 10.5184$$

Problem 3.3.4

Given the following data estimate $f(4.12)$ using Newton-Gregory backward difference interpolation polynomial:

x	0	1	2	3	4	5
$f(x)$	1	2	4	8	16	32

Suppose we have one point $(1, 3)$. How can we find a polynomial that could represent it?

$$P(x) = 3$$
$$P(x) = 3x$$
$$P(1) = 3$$

Suppose we have sequence of points: $(1, 3), (2, 4)$. How can we find a polynomial that could represent it?

$$P(x) = \frac{(x-2)}{(1-2)} \times 3 + \frac{(x-1)}{(2-1)} \times 4$$
$$P(x) = x + 2$$
$$P(1) = 3$$
$$P(2) = 4$$

Suppose we have sequence of points: $(1,3),(2,4),(7,11)$. How can we find a polynomial that could represent it?

$$P(x) = \frac{(x-2)(x-7)}{(1-2)(1-7)} \times 3 + \frac{(x-1)(x-7)}{(2-1)(2-7)} \times 4 + \frac{(x-1)(x-2)}{(7-1)(7-2)} \times 11$$
$$P(1) = 3$$
$$P(2) = 4$$
$$P(7) = 11$$

In a general form it looks like this:

$$P(x) = \frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)}y_1 + \frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}y_2 + \frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)}y_3$$
$$P(x) = \sum_{1}^{3} P_i(x)y_i$$

Definition 3.4.1 (Formula to find the function value)

$$y = \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)}y_0 + \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)}y_1$$

$$+\cdots+\frac{(x-x_1)(x-x_1)\cdots(x-x_{n-1})(x_n-x_0)}{(x_0-x_1)\cdots(x_n-x_{n-1})}y_n$$

$$y = V_o(x_n)y_0 + V_1(x_n)y_1 + V_2(x_n)y_2 + V_3(x_n)y_3 + \cdots + V_n(x_n)y_n$$

Definition 3.4.2 (Find the Polynomial of the given data)

$$P(x) = \sum_{j=0}^{n} y_j \left(\prod_{i=0, i\neq j}^{n} \frac{x-x_i}{x_j-x_i} V_i(x) y_i \right)$$

Example 3.4.3

Using the Lagrange interpolation formula, find the value of y at $x = 0$ given some set of values $(-2, 5)$, $(1, 7)$, $(3, 11)$, $(7, 34)$?

Solution:

$$y(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}y_1$$
$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}y_3$$

Given the known values are, $x = 0; x_0 = -2; x_1 = 1; x_2 = 3; x_3 = 7; y_0 = 5; y_1 = 7; y_2 = 11; y_3 = 34$.

$$y = \frac{(0-1)(0-3)(0-7)}{(-2-1)(-2-3)(-2-7)} \times 5 + \frac{(0-(-2))(0-3)(0-7)}{(1-(-2))(1-3)(1-7)} \times 7$$
$$+ \frac{(0-(-2))(0-1)(0-7)}{(3-(-2))(3-1)(3-7)} \times 11 + \frac{(0-(-2))(0-1)(0-3)}{(7-(-2))(7-1)(7-3)} \times 34$$
$$y = \frac{21}{27} + \frac{49}{6} + \frac{-77}{20} + \frac{51}{54}$$
$$y = \frac{1087}{180}$$

Example 3.4.4

Consider the following table of functional values, find $f(0.06)$ using Lagrange interpolation, where function generated with $f(x) = \log(x)$.

i	x_i	f_i
0	0.40	-0.916291
1	0.50	-0.693147
2	0.70	-0.356675
3	0.80	-0.223144

$$f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}f_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}f_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}f_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}f_3$$

$$y(0.06) = \frac{(0.06-0.50)(0.06-0.70)(0.06-0.80)}{(0.40-0.50)(0.40-0.70)(0.40-0.80)} \times -0.9163$$
$$+ \frac{(0.06-0.40)(0.06-0.70)(0.06-0.80)}{(0.50-0.40)(0.50-0.70)(0.50-0.80)} \times -0.6931$$
$$+ \frac{(0.06-0.40)(0.06-0.50)(0.06-0.80)}{(0.70-0.40)(0.70-0.50)(0.70-0.80)} \times -0.3567$$
$$+ \frac{(0.06-0.40)(0.06-0.50)(0.06-0.70)}{(0.80-0.40)(0.80-0.50)(0.80-0.70)} \times -0.2231$$

$$y(0.06)$$
$$= \frac{(-0.44)(-0.64)(-0.74)}{(-0.10)(-0.30)(-0.40)}(-0.9163) + \frac{(-0.34)(-0.64)(-0.74)}{(0.10)(-0.20)(-0.30)}(-0.6931)$$
$$+ \frac{(-0.34)(-0.44)(-0.74)}{(0.30)(0.20)(-0.10)}(-0.3567) + \frac{(-0.34)(-0.44)(-0.64)}{(0.40)(0.30)(0.10)}(-0.2231)$$

$$
\begin{aligned}
y(0.06) &= \frac{-0.2084}{-0.012} \times -0.9163 + \frac{-0.161}{0.006} \times -0.6931 \\
&\quad + \frac{-0.1107}{-0.006} \times -0.3567 + \frac{-0.0957}{0.012} \times -0.2231 \\
y(0.06) &= -2.11
\end{aligned}
$$

Solution of the polynomial at point 0.06 is $y(0.06) = -2.11$

x_i	f_i	$f[x_i, x_j]$	$f[x_i, x_j, x_k]$	$f[x_i, x_j, x_k, x_l]$
x_0	f_0			
		$f[x_0, x_1] = \frac{f_1 - f_0}{x_1 - x_0}$		
x_1	f_1		$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	
		$f[x_1, x_2] = \frac{f_2 - f_1}{x_2 - x_1}$		$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_2	f_2		$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	
		$f[x_2, x_3] = \frac{f_3 - f_2}{x_3 - x_2}$		
x_3	f_3			

$$
\begin{aligned}
f(x) = f[x_0] &+ (x - x_0) f[x_0, x_1] + (x - x_0)(x - x_1) f[x_0, x_1, x_2] \\
&+ (x - x_0)(x - x_1)(x - x_2) f[x_0, x_1, x_2, x_3]
\end{aligned}
$$

Example 3.5.1

Find solution of $f(x)$ when $x = 151.23855$ using Newton's divided difference interpolation formula for the following data.

x	300	304	305
f(x)	2.4771	2.4829	2.4843

x	y	1st order	2nd order
300	2.4771		
		$\frac{2.4829-2.4771}{304-300}$ =0.0014	
304	2.4829		$\frac{0.0014-0.0014}{305-300}$ =0
		$\frac{2.4843-2.4829}{305-304}$ =0.0014	
305	2.4843		

The value of x at you want to find the $f(x) : x = 151.23855.$
Newton's divided difference interpolation formula is

$$\begin{aligned}
f(x) &= y_0 + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] \\
y(151.23855) &= 2.4771 + (151.23855 - 300) \times 0.0014 \\
&\quad + (151.23855 - 300)(151.23855 - 304) \times 0 \\
y(151.23855) &= 2.4771 + (-148.7614) \times 0.0014 \\
&\quad + (-148.7614)(-152.7614) \times 0 \\
y(151.23855) &= 2.4771 - 0.2083 + 0 \\
y(151.23855) &= 2.2688
\end{aligned}$$

Solution of divided difference interpolation method $y(151.23855) = 2.2688$

x	y	First difference	Second difference	Third difference	Fourth difference
$x_o - 2h$	y_{-2}				
		Δy_{-2}			
$x_o - h$	y_{-1}		$\Delta^2 y_{-2}$		
		Δy_{-1}		$\Delta^3 y_{-2}$	
x_o	y_0		$\Delta^2 y_{-1}$		$\Delta^4 y_{-2}$
		Δy_0		$\Delta^3 y_{-1}$	
$x_o + h$	y_1		$\Delta^2 y_0$		
		Δy_1			
$x_o + h$	y_2				

$$y_p = y_0 + u\left[\frac{\Delta y_{-1} + \Delta y_0}{2}\right] + \frac{u^2}{2!}\Delta^2 y_{-1} + \frac{u\left(u^2-1\right)}{3!}\left[\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right]$$
$$+\frac{u^2\left(u^2-1\right)}{4!}\Delta^4 y_{-2} + \frac{u\left(u^2-1\right)\left(u^2-4\right)}{5!}\left[\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2}\right]$$

Problem 3.6.1

Use Stirling's interpolation formula to find y_{28}, *given that*

$$y_{20} = 48234, y_{25} = 47354, y_{30} = 46267, y_{35} = 44978, y_{40} = 43389.$$

Here the central value is at $x_0 = 30,\ y_0 = 46267$ and step size $h = 5$.
Therefore,

$$u = \frac{28 - 30}{5} = -0.4$$

The difference table is shown below:

x	y_u	Δy_u	$\Delta^2 y_u$	$\Delta^3 y_u$	$\Delta^4 y_u$
$x_{-2} = 20$	$y_{-2} = 48234$				
		$\Delta y_{-2} = -880$			
$x_{-1} = 25$	$y_{-1} = 47354$		$\Delta^2 y_{-2} = -207$		
		$\Delta y_{-1} = -1087$		$\Delta^3 y_{-2} = 5$	
$x_0 = 30$	$y_0 = 46267$		$\Delta^2 y_{-1} = -202$		$\Delta^4 y_{-2} = -103$
		$\Delta y_0 = -1289$		$\Delta^3 y_{-1} = -98$	
$x_1 = 35$	$y_1 = 44978$		$\Delta^2 y_0 = -300$		
		$\Delta y_1 = -1589$			
$x_2 = 40$	$y_2 = 43389$				

The Stirling's interpolation formula is

$$y_u = y_0 + u\left[\frac{\Delta y_0 + \Delta y_{-1}}{2}\right] + \frac{u^2}{2!}\Delta^2 y_{-1} + \frac{u(u^2-1)}{3!}\left[\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2}\right] + \frac{u^2(u^2-1)}{4!}\Delta^4 y_{-2}$$

Substituting the values:

$$y_{28} = 46267 + (-0.4)\left[\frac{-1289 - 1087}{2}\right] + \frac{(-0.4)^2}{2}(-202) + \frac{(-0.4)((-0.4)^2-1)}{6}\left[\frac{-98+5}{2}\right] + \frac{(-0.4)^2((-0.4)^2-1)}{24}(-103)$$

$$y_{28} = 46724.0128$$

Definition 3.7.1

Consider the problem of interpolating between the data points (x_0, y_0), (x_1, y_1), ... (x_n, y_n) by means of spline fitting.
Then the cubic spline $f(x)$ is such that

1. $f(x)$ is a linear polynomial outside the interval (x_0, x_n),
2. $f(x)$ is a cubic polynomial in each of the subintervals,
3. $f'(x)$ and $f''(x)$ are continuous at each point.

Problem 3.7.2

Obtain the cubic spline for the following data:

$x:$	0	1	2	3
$y:$	2	-6	-8	2

Since the points are equispaced with $h = 1$ and $n = 3$, the cubic spline can be determined from $M_{i-1} + 4M_i + M_{i+1} = 6\left(y_{i-1} - 2y_i + y_{i+1}\right), i = 1, 2$.

$$\begin{aligned}
\therefore \qquad M_0 + 4M_1 + M_2 &= 6\left(y_0 - 2y_1 + y_2\right) = 6(2 - 2(-6) - 8) = 36\\
M_1 + 4M_2 + M_3 &= 6\left(y_1 - 2y_2 + y_3\right) = 6(-6 - 2(-8) + 2) = 72\\
i.e., \qquad 4M_1 + M_2 &= 36 \qquad [\because M_0 = 0, M_3 = 0]\\
M_1 + 4M_2 &= 72
\end{aligned}$$

Solving these, we get $M_1 = 4.8, M_2 = 16.8$.

Now the cubic spline in $(x_i \leq x \leq x_{i+1})$ is

$$f(x) = \frac{1}{6}(x_{i+1} - x)^3 M_i + \frac{1}{6}(x - x_i)^3 M_{i+1} + (x_{i+1} - x)\left(y_i - \frac{1}{6}M_i\right) + (x - x_i)\left(y_{i+1} - \frac{1}{6}M_{i+1}\right) \tag{3}$$

Taking $i = 0$ in (3) the cubic spline in $(0 \leq x \leq 1)$ is

$$f(x) = \frac{1}{6}(1 - x)^3(0) + \frac{1}{6}(x - 0)^3(4.8) + (1 - x)(2 - 0) + x\left[-6 - \frac{1}{6}(4.8)\right]$$
$$= 0.8x^3 - 8.8x + 2 \quad (0 \leq x \leq 1)$$

Taking $i = 1$ in (3), the cubic spline in $(1 \le x \le 2)$ is

$$\begin{aligned} f(x) &= \frac{1}{6}(2-x)^3(4.8) + \frac{1}{6}(x-1)^3(16.8) + (2-x)\left[-6 - \frac{1}{6}(4.8)\right] \\ &= 2x^3 - 5.84x^2 - 1.68x + 0.8 \end{aligned}$$

Taking $i = 2$ in (3), the cubic spline in $(2 \le x \le 3)$ is

$$\begin{aligned} f(x) &= \frac{1}{6}(3-x)^3(4.8) + \frac{1}{6}(x-2)^3(0) + (3-x)\left[-8 - \frac{1}{6}(16.8)\right] \\ &\quad + (x-2)\left[2 - \frac{1}{6}(2)\right] \\ &= -0.8x^3 + 2.64x^2 + 9.68x - 14.8 \end{aligned}$$

Problem 3.7.3

The following values of x and y are given :

x:	*1*	*2*	*3*	*4*
y:	*1*	*2*	*5*	*11*

Find the cubic splines and evaluate $y(1.5)$ and $y'(3)$.

Since the points are equispaced with $h = 1$ and $n = 3$, the cubic splines can be obtained from

$$M_{i-1} + 4M_i + M_{i+1} = 6\left(y_{i-1} - 2y_i + y_{i+1}\right), i = 1, 2.$$

$$\therefore \quad M_0 + 4M_1 + M_2 = 6\left(y_0 - 2y_1 + y_2\right)$$

$$M_1 + 4M_2 + M_3 = 6\left(y_1 - 2y_2 + y_3\right)$$

$$i.e., \quad 4M_1 + M_2 = 12,$$

$$M_1 + 4M_2 = 18$$

which give

$$M_1 = 2, M_2 = 4.$$

Now the cubic spline in $(x_i \le x \le x_{i+1})$ is

$$f(x) = \frac{1}{6}\left[(x_{i+1}-x)^3 M_i + (x-x_i)^3 M_{i+1}\right] + (x_{i+1}-x)\left(y_i - \frac{1}{6}M_i\right) + (x-x_i)\left(y_{i+1} - \frac{1}{6}M_{i+1}\right) \quad (4)$$

Thus, taking $i=0, i=1, i=2$ in (4), the cubic splines are

$$f(x) = \begin{cases} \frac{1}{3}\left(x^3-3x^2+5x\right), & 1 \le x \le 2 \\ \frac{1}{3}\left(x^3-3x^2+5x\right), & 2 \le x \le 3 \\ \frac{1}{3}\left(-2x^3+24x^2-76x+81\right), & 3 \le x \le 4 \end{cases}$$

$$f'(x) = \begin{cases} \frac{1}{3}\left(3x^2-6x+5\right), & 1 \le x \le 2 \\ \frac{1}{3}\left(3x^2-6x+5\right), & 2 \le x \le 3 \\ \frac{1}{3}\left(-6x^2+48x-76\right), & 3 \le x \le 4 \end{cases}$$

$$\therefore \qquad y(1.5) = f(1.5) = \frac{11}{8}.$$

Also $y'(3) = \frac{14}{3}$, from both the splines of intervals [2,3] and [3, 4] as they should be.

Problem 3.7.4

Find the cubic spline interpolation for the data:

x :	1	2	3	4	5
$f(x)$:	1	0	1	0	1

Since the points are equispaced with $h = 1, n = 4$, the cubic spline can be found by means of

$$M_{i-1} + 4M_i + M_{i+1} = 6\left(y_{i-1} - 2y_i + y_{i+1}\right), i = 1, 2, 3$$

$$\therefore \quad M_0 + 4M_1 + M_2 = 6\left(y_0 - 2y_1 + y_2\right) = 12$$

$$M_1 + 4M_2 + M_3 = 6\left(y_1 - 2y_2 + y_3\right) = -12$$

$$M_2 + 4M_3 + M_4 = 6\left(y_2 - 2y_3 + y_4\right) = 12$$

Since

$$M_0 = y'' = 0 \text{ and } M_4 = y'' = 0$$

Therefore

$$4M_1 + M_2 = 12; M_1 + 4M_2 + M_3 = -12; M_1 + 4M_3 = 12$$

Solving these equations, we get

$$M_1 = \frac{30}{7}, M_2 = \frac{-36}{7}, M_3 = \frac{30}{7}$$

Now the cubic spline in $(x_i \leq x \leq x_{i+1})$ is

$$f(x) = \frac{1}{6}\left(x_{i+1} - x\right) M_i + \left(x - x_i\right) M_{i+1}\left(x_{i+1} - x\right)\left(y_i - \frac{1}{6}M_i\right) + \left(x - x_i\right)\left(y_{i+1} - \frac{1}{6}M_{i+1}\right) \tag{5}$$

Taking $i = 0$, in (5), the cubic spline in $(1 \leq x \leq 2)$ is

$$\begin{aligned} y &= \frac{1}{6}\left[\left(x_1 - x\right)^3 M_0 + \left(x - x_0\right)^3 M_1\right] + \left(x_1 - x\right)\left(y_0 - \frac{1}{6}M_0\right) \\ &\quad + \left(x - x_0\right)\left(y_1 - \frac{1}{6}M_1\right) \\ &= \frac{1}{6}\left[(2 - x)^3(0) + (x - 1)^3(30/7)\right] + (2 - x)\left[1 - \frac{1}{6}(0)\right] \\ &\quad + (x - 1)\left[0 - \frac{1}{6}\left(\frac{30}{7}\right)\right] \end{aligned}$$

i.e.,

$$y = 0.71x^3 - 2.14x^2 + 0.42x + 2 \quad (1 < x \leq 2)$$

Taking $i = 1$ in (5), the cubic spline in $(2 \leq x \leq 3)$ is

$$y = \frac{1}{6}\left[(3-x)^3\frac{30}{7} + (x-2)^3\left(-\frac{36}{7}\right)\right] + (3-x)\left[0 - \frac{1}{6}\left(\frac{30}{7}\right)\right]$$
$$+ (x-2)\left[1 - \frac{1}{6}\left(-\frac{36}{7}\right)\right]$$

i.e., $y = -1.57x^3 + 11.57x^2 - 27x + 20.28. \quad (2 \leq x \leq 3)$

Taking $i = 2$ in (5), the cubic spline in $(3 \leq x \leq 4)$ is

$$y = \frac{1}{6}\left[(4-x)^3\left(-\frac{36}{7}\right) + (x-3)^3\left(\frac{30}{7}\right)\right] + (4-x)\left[1 - \frac{1}{6}\left(-\frac{36}{7}\right)\right]$$
$$+ (x-3)\left(0 - \frac{5}{7}\right)$$

i.e., $y = 1.57x^3 - 16.71x^2 + 57.86x - 64.57 \quad (3 \leq x \leq 4)$.

Taking $i = 3$ in (5), the cubic spline in $(4 \leq x \leq 5)$ is

$$y = \frac{1}{6}\left[(1-x)^3\left(\frac{30}{7}\right)\right] + (5-x)\left(-\frac{5}{7}\right) + (x-4)(1)$$

i.e., $y = -0.71x^3 + 2.14x^2 - 0.43x - 6.86. \quad (4 \leq x \leq 5)$